

PEDRO H. R. DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

Last updated: May 26, 2026

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – present Rio de Janeiro, Brazil

Technical Degree in Informatics

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
February. 2020 – December. 2023 Rio de Janeiro, Brazil

RESEARCH PROJECTS

Understanding Dark Matter from ExtraGalactic Shocks

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
September. 2022 – March. 2023 Rio de Janeiro, Brazil

Junior Scientific Initiation Scholarship - CNPq
Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – September. 2023 Rio de Janeiro, Brazil

Volunteer in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – February. 2024 Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - IFF
Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning 2025

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
October. 2025 – present Rio de Janeiro, Brazil

Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2020 Rio de Janeiro, Brazil

Operational Systems

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2021 Rio de Janeiro, Brazil

Server Administration

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2022 Rio de Janeiro, Brazil

RESEARCH INTERESTS

Stars Milky Way Galactic Archaeology Star Evolution Quantum Mechanics Computational Physics ML AI
Physics Astrophysics Computational Deduction of Equations Animations Simulations

EVENTS

Science Fairs

21st Brazilian Science and Engineering Fair

Poster [7]
USP, São Paulo

March. 2023 São Paulo, Brazil

40th International Science and Technology Fair

Poster [9]
Fundação Librelato, Novo Hamburgo

October. 2023 Rio Grande do Sul, Brazil

IX Brazilian Science Initiation Fair

Poster [6]
Pomerode

September. 2024 Santa Catarina, Brazil

XVI Scientific and Technological Exhibition of the Açaí Institute

Poster [4]
UFPA, Belém

December. 2025 Pará, Brazil

Conferences

44° CONUBES

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Arena Nilson Nelson, Brasília

May. 2022 Distrito Federal, Brazil

XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies

Oral Presentation [5]
UENF, Campos dos Goytacazes

May. 2024 Rio de Janeiro, Brazil

XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies

Oral Presentation [3]
UENF, Campos dos Goytacazes

May. 2025 Rio de Janeiro, Brazil

XLVIII Annual Meeting of the Brazilian Astronomical Society

Poster [1, 11]
Hotel Glória, Caxambu

September. 2025 Minas Gerais, Brazil

XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies

Talk [2]
UENF, Campos dos Goytacazes

May. 2026 Rio de Janeiro, Brazil

Academic Weeks

VIII Knowledge Showcase and I Job Fair

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

December. 2020 Rio de Janeiro, Brazil

IX Knowledge Showcase and II Job Fair

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

October. 2021 Rio de Janeiro, Brazil

I Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

May. 2021 Rio de Janeiro, Brazil

X Knowledge Showcase and III Job Fair

Talk
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

October. 2022 Rio de Janeiro, Brazil

II Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

May. 2022 Rio de Janeiro, Brazil

XI Knowledge Showcase and IV Job Fair

Poster [8, 10]
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

September. 2023 Rio de Janeiro, Brazil

III Week of Computer Engineering and Management

Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*

 PUBLICATIONS

 Link Nasa ADS: [NASA ADS - Pedro H. R. de Andrade](#)

Proceedings

- [1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. “Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4”. In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.
- [2] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. “Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4”. In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.
- [3] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecília Soja. “Simulando o impacto de satélites em observações astronômicas”. In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.
- [4] Maycon Jorge Deláqua da Silva et al. “Simulando o impacto de satélites em observações astronômicas”. In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.
- [5] Maycon Jorge Deláqua da Silva et al. “Simulando o impacto de satélites em observações astronômicas”. In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.
- [6] Maycon Jorge Deláqua da Silva et al. “Simulando o impacto de satélites em observações astronômicas”. In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.
- [7] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. “Análise de simulações dinâmicas de aglomerados de galáxias em fusão”. In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.
- [8] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. “Entendendo a matéria escura a partir de choques extragaláticos”. In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 17.
- [9] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Fabrício Barros Gonçalves. “Entendendo a matéria escura a partir de choques extragaláticos”. In: *Anais da MOSTRATEC 2023*. 2023, p. 102.
- [10] Gabriel Coelho de Souza Poubel et al. “MobFog no IFFMaker”. In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

- [11] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. “Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4”. In: *arXiv e-prints* (Dec. 2025). DOI: [10.48550/arXiv.2512.10037](#).

 AWARDS

-  IX FEBIC
3rd Place in the Undergraduate Category with the work “Simulating the Impact of Satellites on Astronomical Observations”
-  XVI MCTIA
1st Place in the Undergraduate Category with the work “Simulating the Impact of Satellites on Astronomical Observations”

 STATISTICS

-  3 Oral Presentations
National Congresses
-  7 Posters
National and International Events
-  9 Science Fairs/Congresses
National and International Events
-  10 Academic Weeks
Regional Events

 PROGRAMMING

-  Python
-  git Git
-  Github
-  L^AT_EX
-  C++
-  Java
-  NodeJS

 LANGUAGES

- Portuguese
- English
- Francês
- Spanish